

CHAPTER 7

RESULTS, DISCUSSION, CONCLUSION AND FUTURE SCOPE

7.1 RESULTS AND DISCUSSION

The overarching objective of this project was to eliminate the reliance on subjective manual visual judgment by providing a continuous, objective risk assessment of bus stop overcrowding. The development, testing, and deployment phases—detailed across Chapters 4, 5, and 6—confirmed that the integrated IoT and machine learning architecture successfully achieved this goal. Synthesizing the technical findings with the community impact assessment reveals a highly reliable, low-cost system that successfully bridges the gap between edge-level physical sensing and cloud-based predictive analytics.

From a technical hardware perspective, the system demonstrated exceptional stability. Testing results from Chapter 5 yielded a 100% pass rate across all 11 unit tests (UT-01 to UT-11) and all 6 integration tests (IT-01 to IT-06). The ESP32 edge node effectively managed the physical sensing layer, confirming passenger crossings with a highly responsive 150 ms polling interval. By utilizing a strict logic filter that required two consecutive readings below 400 mm, the hardware successfully rejected transient sensor noise and prevented false crossing registrations. During early development, three critical bugs were systematically identified and resolved: a VL53L0X library API mismatch, a VN versus 3.3 V power rail miswiring that left sensors unpowered, and a V2/V3 virtual pin ownership conflict. Resolving these resulted in a finalized 3.3 V architecture and strict pin-ownership protocol that performed flawlessly during continuous deployment.

The performance of the backend processing and communication layers was equally robust. The Python backend (brain.py) maintained a strict 5-second processing cycle, ensuring the system remained highly responsive to real-time microclimate and crowding changes. The communication bridge via Blynk Cloud recorded an average round trip latency of 891 ms, well within acceptable operational limits. Crucially, the Telegram alerts were delivered to the transit authority in just 0.81 seconds. To ensure the system preserved a persistent historical record for future analysis and retraining, operational data was appended to the local bus_stop_log.csv file in effectively 0.00 ms per cycle.

The most significant technical finding pertained to the machine learning model's performance and its precise alignment with the foundational risk formula. The RandomForestRegressor, configured with 100 trees and a random_state=42, executed its risk computation in just 10.51 ms, making it exceptionally well-suited for a live IoT pipeline. Because the model was trained on a dense, formula-generated synthetic grid of 43,632 samples (covering 24 hours, 101 passenger counts, 2 rain states, and 9 temperatures), it perfectly approximated the non-linear environmental penalties of the system without overfitting. The model output exactly matched the expected

formula results under the tested scenarios: predicting a 90.0% CRITICAL risk at full capacity (45 passengers) with rain, a 65.0% WARNING risk for 30 passengers with rain, and a baseline 15.0% NORMAL risk for 0 passengers with rain.

From an operational and community perspective, the system performed exactly as intended during the simulated pilot deployment at the Dindoshi Bus Depot (Route 326) context. The strict boundary logic—where a score of exactly 75 remains classified as WARNING, and only a score above 75 triggers CRITICAL—was pivotal in preventing false-positive emergencies. Overcrowding alone (reaching 75) was intentionally not treated as an emergency; environmental pressure (rain, heat $\geq 33^{\circ}\text{C}$, or rush hour) was required to cross the critical threshold. When this threshold was breached, the community impact was immediate: on-site commuters were informed via the red LED indicator (with public audio firmware-disabled for safety), while remote transit authorities received automated Telegram alerts. The enforcement of a 60-second alert cooldown successfully prevented notification fatigue for the operators.

Finally, the V7 "Bus Left (-30)" push button and the V8 "minutes since last crossing" display proved to be a highly effective human-in-the-loop design decision. Instead of utilizing an automatic count decay—which risks erasing waiting commuters if a bus reaches capacity and leaves stragglers behind—the system placed the final subtractive control in the hands of the human operator. By referencing the V8 inactivity timer as concrete evidence before pressing V7, transit staff maintained safe, contextual control over the system's tracking accuracy.

7.2 ACHIEVEMENT OF OBJECTIVES

The project's original objectives, defined in Chapter 1, were systematically evaluated against the final built prototype to determine the extent of their completion. The results confirm that all primary objectives were successfully fulfilled.

Table 7.1: Status of Project Objectives

Objective	Status	Evidence
1. To design a low-cost IoT hardware node for counting passengers and monitoring ambient temperature at a single bus stop.	Met	The ESP32 successfully integrates the VL53L0X (passenger crossings via GPIO21/22) and DHT22 (temperature via GPIO4) on a unified 3.3 V power rail.
2. To integrate live environmental context (weather and time) with local sensor data.	Met	The Python backend successfully fetches live Dindoshi weather via the OpenWeatherMap API, mapping WMO codes to a binary rain state.
3. To develop a Machine Learning model capable of predicting a continuous crowding risk score.	Met	A RandomForestRegressor trained on 43,632 samples accurately outputs a 0–98% continuous risk score matching the raw risk formula.
4. To provide transparent, real-time	Met	Local LEDs (GPIO25/26) provide on-

alerts to both on-site commuters and remote transit authorities.		site awareness, while Telegram Bot API alerts trigger perfectly under the above 75 strict CRITICAL rule.
5. To allow safe operator correction of passenger counts following a bus departure.	Met	The V7 "Bus Left" push button and V8 "minutes since last crossing" evidence gauge were successfully implemented on the Blynk dashboard.

7.3 PROJECT CONTRIBUTIONS

This project introduces a highly accessible, stop-level predictive tool that fills a significant technical and operational gap in current public transit management. Previously, transit authorities relied on expensive, vehicle-mounted Industrial Automatic Passenger Counting (APC) systems, which measure fleet load but ignore the physical conditions at the shelter. Alternatively, commuters utilized transit applications like Chalo or Google Maps, which track vehicle GPS locations but remain entirely blind to the actual crowding pressure at a specific stop.

The primary technical contribution of this project is the successful fusion of anonymous, edge-level Time-of-Flight (ToF) sensing with live cloud-based microclimate data and a Random Forest machine learning backend, all achieved using exceptionally low-cost hardware. By translating the feature set [hour_of_day, is_raining, passenger_count, temperature] into a transparent, actionable risk percentage, this approach replaces subjective visual judgment with objective, quantifiable data.

The practical contribution to the community is the establishment of a proactive safety measure. Commuters no longer have to guess the severity of the crowding, and operators no longer have to wait for visual confirmation or commuter complaints to dispatch feeder buses. Beyond the pilot users at the Dindoshi Bus Depot, this architecture holds broader relevance for municipal bodies, such as the Mumbai Metropolitan Region Development Authority (MMRDA), and suburban transit networks. It provides a scalable blueprint for monitoring unstaffed, high-density pedestrian infrastructure without investing in invasive CCTV cameras or expensive fleet-wide technology upgrades.

7.4 LIMITATIONS

While the prototype successfully meets its core requirements and functional specifications, several practical and technical limitations were identified during the development, testing, and deployment phases.

- **Single Sensor Ghost Counting:** Because the system utilizes a single VL53L0X Time-of-Flight sensor, it can only detect proximity crossings; it cannot explicitly calculate entry versus exit direction. An individual lingering or pacing within the sensor's 400 mm field of

view could trigger ghost counts, artificially inflating the passenger tally. This hardware constraint makes the V7 operator correction essential for maintaining long-term accuracy.

- **Estimated Capacity Constraint:** The maximum capacity baseline of 45 passengers is a reasoned design estimate, not a physically measured metric for the specific Dindoshi shelter. It was calculated by estimating a typical shelter footprint of 36 m² and dividing it by 0.80 m² per passenger, which strictly aligns with the Level of Service C (LOS C) comfortable standing density standard defined by the transit-design Highway Capacity Manual.
- **Synthetic Training Data:** The Random Forest model was trained on a 43,632-sample formula-generated grid rather than historical, real-world crowding data. Because localized sensor-fusion data for this specific stop did not previously exist, the model currently acts as a highly accurate formula approximator rather than an engine discovering hidden real-world patterns.
- **Wi-Fi Dependency:** The backend predictive logic and alerting mechanisms depend entirely on continuous Wi-Fi connectivity. If the internet connection drops, the ESP32 edge node will successfully continue counting passengers locally, but the live weather fetching, risk computation, Telegram alerts, and Blynk dashboard updates will fail until the network connection is restored.

7.5 FUTURE ENHANCEMENTS

The current prototype establishes a robust, foundational architecture that can be significantly expanded and improved through several targeted future enhancements.

- **Directional Sensing Integration:** Upgrading the hardware edge layer to utilize dual VL53L0X sensors, or transitioning to a multi-zone Time-of-Flight sensor such as the VL53L5CX, would allow the microcontroller to calculate entry and exit vectors. This would effectively solve the ghost-counting limitation, creating an autonomous, self-correcting tally that reduces the reliance on manual operator corrections.
- **Real-World Model Retraining:** The system currently appends the results of every 5-second processing cycle to a local bus_stop_log.csv file. Over time, this will generate a massive dataset of authentic commuter behavior at the Dindoshi stop. In the future, this accumulated CSV data can be used to retrain the Random Forest model, shifting it from a synthetic formula-approximation tool to an empirical, data-driven predictive model based on true historical trends.
- **Cloud Database Migration:** Transitioning the persistent storage architecture from a localized CSV file on a single backend machine to a managed cloud database (such as MySQL or Firebase) would allow for complex historical querying. This migration is a necessary step for supporting a scalable architecture that spans multiple bus stops and transit routes across the city.

- **Solar Power Implementation:** Replacing the current USB power dependency with a localized solar panel and battery management system would allow the ESP32 edge node to be deployed entirely off-grid, expanding the system's viability to remote or underdeveloped transit shelters.

7.6 CONCLUSION

This project set out to resolve the lack of objective, real-time overcrowding awareness at public bus stops by replacing reactive, manual visual judgment with an automated predictive system. By integrating a low-cost ESP32 controller, an anonymous VL53L0X Time-of-Flight sensor, and a DHT22 module, the hardware successfully captured localized physical conditions at the edge. When fused with live OpenWeatherMap data and processed through a RandomForestRegressor machine learning model, the system generated highly accurate, continuous risk scores that dynamically accounted for rain, heat, and rush-hour pressures.

The technical execution proved highly reliable, characterized by 100% test pass rates, rapid 150 ms sensor polling, and 10.51 ms predictive computations. The deployment impact on the community was immediate and highly practical: commuters gained on-site situational awareness via glanceable LED indicators, while transit authorities received automated Telegram alerts the moment the risk crossed the strict above 75 CRITICAL threshold. Ultimately, this IoT-based solution proves that scalable, low-cost sensor fusion can proactively safeguard commuter transit networks, providing vital overcrowding intelligence without compromising public privacy or requiring massive infrastructural investments in CCTV or industrial APC systems.